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# Informative Model Validation for Anomaly Detection

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## Abstract

Model selection and model validation for anomaly detection are difficult problems. Labelled anomalies are usually unavailable and synthetic anomalies introduce bias. We propose approximation capacity (CAP), a signal-agnostic criterion derived from posterior agreement. This criterion scores a model by measuring both the consistency and the confidence of its predictions across two background datasets. CAP is evaluated across three data domains and several anomaly detector models. We find that our criterion is competitive with stability and distribution-matching validation approaches, often outperforming them. Thus, CAP provides a practical alternative to both synthetic-signal and stability-only model validation.

## 1 Introduction

Anomaly detection (AD) is a machine learning (ML) task pertaining to the differentiation of unusual data from *typical* data. Anomaly detectors commonly undergo *unsupervised* training on a dataset that consists predominantly of typical samples, without any significant anomalous data contamination. The success of this paradigm rests on the following conjecture: if the model learns to describe *typicality* well, then it is able to discern anomalies as deviations from the typicality description [1–6]. This characterises the standard ML anomaly detector, which appears in many domains, from industrial applications [7–10] to particle physics [11–13], gravitational waves [14–16], and medical uses [17]. The paradigm described hitherto became standard because labelled anomaly datasets are extremely difficult to construct in real-world scenarios. Anomalous events are inherently rare, diverse, and often unknown *a priori*. Consequently, model selection and validation are fundamentally challenging.

The model selection and validation of an anomaly detector are disambiguated and defined as follows. Model selection refers to choosing a specific checkpoint along the training trajectory. Model validation pertains to evaluating performance across different hyperparameter configurations. The former determines a specific model instance while the latter determines model capacity and training dynamics. The common approach to both involves performance estimation on synthetic anomaly data [18, 19]. However, this biases the validation process towards peculiarities in the synthetic dataset [5, 20]. Another common approach uses model stability metrics on typical data as proxies for anomaly detection performance [10, 21, 22]. Nevertheless, such proxies do not directly capture informativeness: a stable model may correspond either to a performant detector or to a trivial solution.

We propose a new selection and validation criterion that is both *informative* and *signal-agnostic*. This criterion is based on the posterior agreement (PA) framework [23], which favours models whose probabilistic predictions are consistent across independently sampled data, while remaining confident. Thus, we derive the *approximation capacity* (CAP) metric by adapting the general PA formulation to the context of anomaly detection. Like PA, CAP characterises the robustness of a model by assessing the consistency of its anomaly scores across typical datasets, while penalising low-confidence scoring. This is achieved without the use of labelled anomalies. The performance of our new selection and validation paradigm is highlighted in comparison with standard AD model validation approaches, across several anomaly detector architectures and different data modalities.

39 **Contributions** The main contributions of this work are summarised as follows:

- 40 • Adapting the standard posterior agreement framework to the anomaly detection application:  
41 deriving the approximation capacity metric, implementing it, and calibrating it.
- 42 • A principled model selection and validation pipeline that enables a fair comparison between  
43 CAP and conventional anomaly detection validation strategies.
- 44 • CAP is evaluated on three anomaly-detection tasks and several detector architectures, and  
45 is shown to be a competitive signal-agnostic validation criterion that often outperforms  
46 standard signal-agnostic baselines.

## 47 2 Related work

48 **Anomaly detectors.** Anomaly detection methods aim to identify deviations from a learned rep-  
49 resentation of normality, and differ primarily in how this typical data representation is constructed.  
50 Classical AD approaches formulate anomaly detection as a one-class classification problem, where  
51 a boundary enclosing the typical data is learned [1]. One-class anomaly detectors include support  
52 vector machines (SVMs) [2], support vector data description (SVDD) models [3, 4, 18], kernel  
53 density algorithms [24], and recent energy-based models [25, 26]. Reconstruction methods model  
54 typicality by learning to reconstruct inputs: anomalies are detected via large reconstruction errors.  
55 Autoencoder-based approaches form the basis of this latter paradigm and have been widely adopted  
56 in anomaly detection [5]. These methods are particularly flexible and have been used across domains,  
57 including industry-related applications of anomaly detection and scientific applications involving  
58 high-dimensional detector data [9, 12]. Density-based methods explicitly estimate the data distri-  
59 bution, assigning low likelihood to anomalous samples, for instance using normalizing flows [27].  
60 Variational autoencoders provide an alternative approach by learning approximate likelihoods through  
61 latent variable models [28]. More recent approaches leverage self-supervised and representation  
62 learning techniques to learn features that separate typical data from anomalous data [10, 19, 29].  
63 All of these methods rely on anomaly scoring functions defined with respect to typical data.

64 **Model validation in AD.** The absence of labeled anomaly datasets makes model validation for  
65 anomaly detection a challenging problem. A common approach relies on performance estimation  
66 using labeled or synthetic anomalies, as in supervised or semi-supervised AD settings [18, 19].  
67 However, such a paradigm introduces biases: the selected models tend to learn specific features of  
68 the chosen anomaly distribution, which may not be representative of real-world anomalies [5, 20].  
69 Alternatively, model validation can be guided by proxy metrics computed on typical data, such as  
70 reconstruction error, likelihood estimates, or criteria that capture anomaly assignment stability [10, 22].  
71 While these proxies can provide useful guidance, they do not directly measure the ability of a model to  
72 distinguish anomalies from typical data, and may result in selecting trivial or uninformative solutions.

73 **Information-theoretic learning.** Information-theoretic methods are commonly used in machine  
74 learning to quantify the quality of learned representations. An early example is Tishby’s information  
75 bottleneck principle [30]. More recently, mutual information criteria have seen uses in contrastive  
76 learning methods [31, 32]. The main aim of such methods is to capture informative structure by  
77 preserving relevant information while discarding noise.

78 In this context, the posterior agreement framework provides a principled approach to model selection  
79 by quantifying the consistency of an algorithm’s probabilistic predictions across datasets [23, 33, 34].  
80 Posterior agreement defines a Gibbs distribution over model hypotheses and evaluates their agreement  
81 across different datasets drawn from the same underlying process. Thus, model validation based on  
82 posterior agreement balances consistency with the entropy of the induced anomaly score distribution.  
83 This formulation naturally captures a trade-off between robustness and informativeness: models that  
84 produce consistent yet confident predictions across datasets are preferred.

85 Recent work shows that posterior agreement can be used to select models based on the consistency of  
86 their predictions in adversarially perturbed settings [35]. Models are preferred when they produce  
87 similar predictions under such variations in the data. The framework also penalizes uninformative  
88 or degenerate outputs, such as uniform predictions that lack discriminative power. This prevents  
89 selecting models that are stable but fail to capture meaningful structure. Such a criterion is well suited  
90 for signal-agnostic model selection and validation when labelled anomalies are unavailable.

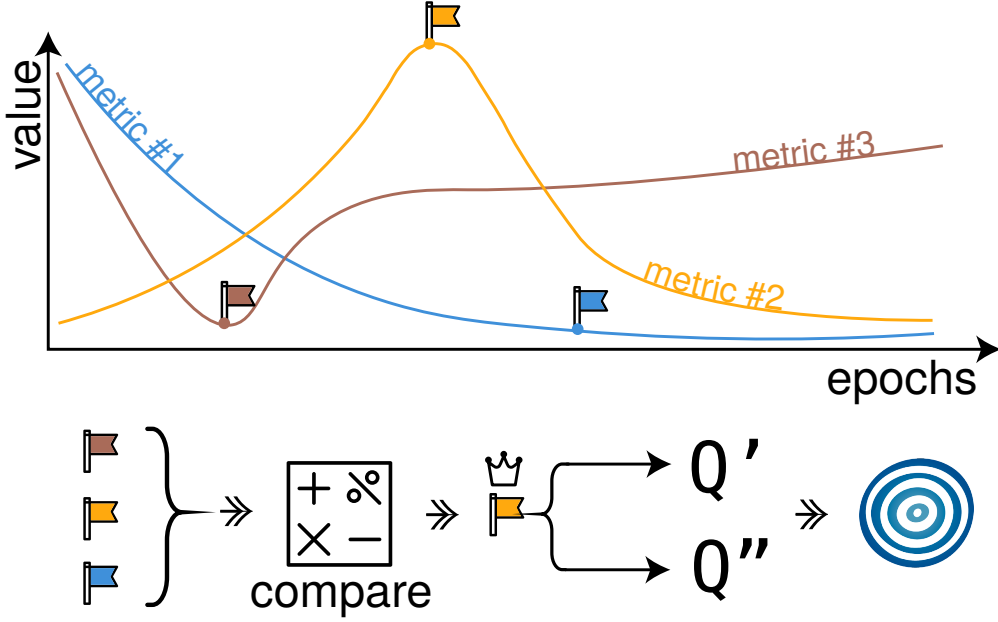


Figure 1: The model selection and validation pipeline design. At the top, the training of a trial is illustrated together with three different checkpointing strategies, represented by the flags. The bottom of the plot shows the checkpoint selection mechanism implemented in this work. The checkpoints are compared using a fixed rule on  $Q'$  and  $Q''$ . The primary validation quantity  $Q'$  is particular to the framework under study: for example, threshold drift for the stability approach or CAP for our method. Meanwhile,  $Q''$  is the model’s native anomaly-score criterion, e.g., the mean squared error for an autoencoder. The winning checkpoint passes these quantities to Optuna, which registers a trial.

### 91 3 Methodology

92 The proposed model selection and validation pipeline is shown in Fig. 1. For every hyperparameter  
 93 trial, we train a model and save a set of candidate checkpoints according to a checkpointing strategy.  
 94 Then, each checkpoint is evaluated on two validation quantities. The *primary* quantity,  $Q'$ , is the  
 95 validation principle under study: the semi-supervised metric, stability, Wasserstein distance, or CAP.  
 96 The *secondary* quantity,  $Q''$ , is the model’s native anomaly-score criterion, such as the reconstruction  
 97 loss or the KL divergence, averaged above a certain threshold. These two quantities define a 2D  
 98 objective space from which we select a single checkpoint by applying a fixed comparison rule.

99 The fixed comparison rule is defined as follows. Let the pair  $(Q'_c, Q''_c)$  correspond to the currently  
 100 best checkpoint and  $(Q'_n, Q''_n)$  be the contender checkpoint. For comparing  $Q'$ , we define

$$\Delta_{Q'} = \begin{cases} \frac{Q'_n - Q'_c}{\max(|Q'_c|, \varepsilon)}, & \text{if the primary metric is maximized,} \\ \frac{Q'_c - Q'_n}{\max(|Q'_c|, \varepsilon)}, & \text{if the primary metric is minimized,} \end{cases} \quad (1)$$

101 and analogously  $\Delta_{Q''}$  for the secondary quantity, where  $\varepsilon = 10^{-12}$  to ensure numerical stability. We  
 102 then form the weighted score  $S = 1.25 \Delta_{Q'} + \Delta_{Q''}$ . This gives a mild fixed preference to the primary  
 103 quantity  $Q'$  while retaining the regularising effect of  $Q''$ ; it does not alter their commensurability.  
 104 Because both terms are relative changes, they are dimensionless. The new checkpoint replaces the  
 105 current one if  $S > 0$ . The first candidate checkpoint is accepted automatically: it does not need to be  
 106 better than a reference. The order of comparison with subsequent checkpoints is fixed. Applying  
 107 this procedure to all trials yields one checkpoint per trial, which can then be compared across  
 108 hyperparameter configurations. For the selected checkpoint, we report  $(Q', Q'')$  to a hyperparameter  
 109 optimization algorithm of choice. In the following subsections,  $Q'$  and  $Q''$  are defined for the four  
 110 model selection and validation frameworks explored in this work.

### 3.1 Conventional model validation criteria

**Semi-supervised.** The semi-supervised strategy serves as an upper-bound reference. Its primary quantity  $Q'$  is a labeled downstream metric used for model selection and validation. Here,  $Q'$  is the average of the worst-performing 25% of downstream efficiencies across the synthetic anomaly datasets, i.e., a lower-tail conditional value-at-risk metric denoted  $\text{CVaR}_{0.25}$ . Let  $r_1, \dots, r_n$  be the downstream efficiencies on the anomaly datasets, and let  $r_{(1)} \leq \dots \leq r_{(n)}$  be the sorted values. For a model checkpoint  $\theta$  under evaluation, we define

$$\text{CVaR}_{0.25}(\theta) = \frac{1}{k} \sum_{i=1}^k r_{(i)}, \quad k = \lceil 0.25 n \rceil, \quad (2)$$

which is maximized. The  $\text{CVaR}_{0.10}$  is also considered; the results do not change qualitatively. The secondary quantity  $Q''$  is the model's native anomaly score criterion on a validation dataset comprised of typical samples. This strategy is effective, but biased by construction, since it uses signal information unavailable in a true anomaly discovery setting.

**Stability-based.** The stability strategy is fully signal-agnostic. Its primary quantity  $Q'$  is a split-transfer threshold-drift metric. Let  $B = \{r_i\}_{i=1}^N$  be the anomaly scores on the validation split of the typical sample dataset that was used for training. Additionally, let  $B_A$  and  $B_E$  denote a deterministic internal split into calibration and evaluation subsets, with sizes  $N_A$  and  $N_E$ . For a model  $\theta$  under evaluation, a threshold is computed on the calibration split as the empirical  $(1 - \text{FPR})$ -quantile

$$\tau_A(\theta) = Q_{1-\text{FPR}}(\{f_\theta(x) : x \in B_A\}), \quad (3)$$

where  $f_\theta(x)$  is the anomaly score for checkpoint  $\theta$  and FPR is the false positive rate of anomalies. The transferred exceedance rate on the evaluation split is

$$\hat{p}_E(\theta) = \frac{1}{N_E} \sum_{x \in B_E} \mathbf{1}[f_\theta(x) \geq \tau_A(\theta)]. \quad (4)$$

With the finite-sample correction  $\varepsilon = 0.5/N_E$ , the primary metric  $Q'$  for the stability criterion is

$$L_{\text{thr}}(\theta) = \left| \log \frac{\hat{p}_E(\theta) + \varepsilon}{\text{FPR} + \varepsilon} \right|. \quad (5)$$

which is minimized in this paradigm. Intuitively, Eq. 5 measures how much the transferred exceedance rate deviates from the target rate on a logarithmic scale. Similar to the supervised case, the secondary quantity  $Q''$  for this model selection and validation framework is the model's native anomaly score criterion on the validation split of typical data. This criterion favours thresholds that transfer stably across typical data validation splits, but can still select models and hyperparameters that are stable because they give trivial assignments.

**Distribution-similarity.** The distribution-similarity strategy is likewise signal-agnostic. Its primary quantity is the Wasserstein distance between anomaly-score distributions on two validation splits that represent typical data. Let  $R^{(1)} = \{r_i^{(1)}\}_{i=1}^{n_1}$  and  $R^{(2)} = \{r_j^{(2)}\}_{j=1}^{n_2}$  be the corresponding scalar anomaly scores. We first apply the monotone transform  $\tilde{r} = \log(1 + r)$  on the anomaly scores to increase robustness against outliers. For a checkpoint  $\theta$  under evaluation, the primary metric  $Q'$  is the Wasserstein-1 (W1) distance

$$W_1(\theta) = \int_0^1 |q_{\tilde{R}^{(1)}}(u) - q_{\tilde{R}^{(2)}}(u)| du, \quad (6)$$

where  $q_{\tilde{R}^{(1)}}$  and  $q_{\tilde{R}^{(2)}}$  are empirical quantile functions. In the equal-sample case, this reduces to

$$W_1(\theta) = \frac{1}{n} \sum_{i=1}^n \left| \tilde{r}_{(i)}^{(1)} - \tilde{r}_{(i)}^{(2)} \right|, \quad (7)$$

which is minimised in this paradigm. As for the previous two frameworks, the secondary quantity is the model's native anomaly score criterion on the typical sample validation dataset. This baseline captures marginal distribution similarity across sources, but not whether the two sources support the same latent anomaly and typical data assignments.

### 3.2 Approximation Capacity

We derive the approximation capacity (CAP) metric: an adaptation of the general posterior agreement framework [23, 33, 34] to the problem of signal-agnostic model validation in anomaly detection. The goal of this metric is to compare two datasets that share the same notion of typicality, but are not naturally aligned sample by sample. This task-specific formulation of posterior agreement is defined by four choices: i) a binary latent hypothesis space corresponding to typical and anomalous behaviour, ii)  $n(x)$  a normalization function for the anomaly scores, iii)  $P(X_1, X_2)$  a sample to sample pairing function for unaligned datasets, and iv)  $E(p, y)$  an energy function for the posterior agreement kernel. Together, these choices define the exact CAP instantiation used throughout this work.

First, we show how the general posterior agreement kernel specialises to the anomaly detection task. Consider the posterior agreement kernel as defined in Ref. [35]:

$$k_{\text{PA}}(X_1, X_2) = \log \left( |C| \prod_{i=1}^N \sum_{j=1}^K p_i(j | X_1) p_i(j | X_2) \right), \quad (8)$$

where  $X_1$  and  $X_2$  denote the two datasets being compared,  $C$  is the hypothesis space, and  $|C|$  is its cardinality,  $N$  is the number of paired observations from the two datasets,  $K$  is the number of latent hypotheses, and  $p_i(j | X_q)$  is the posterior probability, under dataset  $X_q$ , that the  $i$ -th observation is assigned to latent class  $j$ . In the anomaly detection setting, the latent hypothesis space is binary, with  $y \in \{0, 1\}$  corresponding to typical data and anomaly, so  $K = 2$  and Eq. 8 becomes

$$k_{\text{PA}}(X_1, X_2) = \log |C| + \sum_{i=1}^N k(x_1^{(i)}, x_2^{(i)}), \quad (9)$$

where the  $\log |C|$  term is dropped because it does not affect the ordering induced by CAP itself. Because the checkpoint rule uses relative improvements, this constant shift rescales  $\Delta_{Q'}$ ; dropping it therefore fixes the CAP scale used in the combined score. Further, a Gibbs distribution  $e^{-\beta E(p, y)}$  is assigned to each paired score  $(p_i^{(1)}, p_i^{(2)})$  [23] to obtain

$$k(x_1^{(i)}, x_2^{(i)}; \beta) = \log \left( \frac{e^{-\beta(E(p_i^{(1)}, 0) + E(p_i^{(2)}, 0))} + e^{-\beta(E(p_i^{(1)}, 1) + E(p_i^{(2)}, 1))}}{z_i^{(1)} z_i^{(2)}} \right) \equiv k_i(\beta), \quad (10)$$

with

$$z_i^{(1)} = e^{-\beta E(p_i^{(1)}, 0)} + e^{-\beta E(p_i^{(1)}, 1)}, \quad z_i^{(2)} = e^{-\beta E(p_i^{(2)}, 0)} + e^{-\beta E(p_i^{(2)}, 1)}, \quad (11)$$

the corresponding partition functions and where  $\beta$  is the inverse temperature parameter, which is optimised using gradient descent whenever the metric is computed. Following assumption i), we derived the anomaly detection posterior agreement kernel in Eq. 9. Thus, CAP is defined as

$$\text{CAP}(\theta) = \max_{\beta \geq 0} \frac{1}{N} \sum_{i=1}^N k_i(\beta), \quad (12)$$

where  $\theta$  denotes the model under evaluation. The numerator of  $k_i(\beta)$  in Eq. 10 contains only the agreement states (0, 0) and (1, 1), while the denominator is the fully factorized hypothesis mass. Hence,  $k_i(\beta) \leq 0$ , where values closer to 0 indicate stronger agreement. If both sources are maximally ambiguous, the posterior mass is spread evenly over the two latent labels and the kernel approaches the uninformative baseline of  $-\log 2$ . CAP therefore rewards *agreement with confidence*, not agreement alone, in contrast with the distribution-similarity method from Sec. 3.1.

Choices ii)-iv) are specific to the task under consideration and hence these are determined empirically. The most stable CAP results are obtained by using iii) empirical CDF pairing, then normalising the paired distributions by passing them through ii) a sigmoid function, and finally computing iv) the energy function for the CAP kernel:

$$E(p, y) = y[(1 - p) + \max((\mu + \lambda\sigma) - p, 0)] + (1 - y)[p + \max(p - (\mu - \lambda\sigma), 0)], \quad (13)$$

which we call the adaptive energy, where  $\mu$  and  $\sigma$  denote the mean and standard deviation of the paired normalized scores, and  $\lambda \equiv 0.5$  is a scale hyperparameter. For the study that is conducted to determine choices ii)-iv), which leads to the exact instantiation of CAP used in this work, see App. A.

CAP( $\theta$ ) is maximized in this paradigm. It is the primary quantity  $Q'$  in the new model validation and selection strategy proposed in this work. The secondary quantity is still the model's native anomaly score criterion determined on the typical sample validation split.

## 4 Datasets

**LHC L1 AD.** Our main application is unsupervised anomaly detection in proton–proton collision data collected with the CMS experiment at CERN. This setting is fundamentally signal-agnostic: assuming prior knowledge about the detector signature of a new particle is an impossible counterfactual. Thus, using simulated anomalies during validation or model selection would introduce an undesirable inductive bias toward the chosen signal families. For signal-agnostic validation, this dataset contains real typical data and a simulation of this typical data, which are assumed to share the same underlying notion of normality. Only the real typical data is used for training. Meanwhile, typical and simulated typical samples are used in computing the CAP and Wasserstein metrics. Furthermore, 19 simulated anomaly datasets are reserved strictly for downstream evaluation. Performance is studied at the extreme operational quantile  $q \approx 0.999991$ , corresponding to the 250 Hz regime relevant at the LHC. Further details of the dataset construction and preprocessing are deferred to App. B.

**CIFAR-10.** To complement the scientific setting with a standard image benchmark, we also evaluate on CIFAR-10 [36]. We use class 0 as the selected typical class and cast anomaly detection as distinguishing this class from the remaining classes. For signal-agnostic validation, the two typical domains are obtained by splitting class 0 into two disjoint subsets, equal in samples, so that CAP and Wasserstein compare two views of the same notion of typicality. One of the typical subsets is split further into 80% training data and 20% validation data; the other is used solely for testing. The nine remaining CIFAR-10 image classes are then used strictly for downstream anomaly evaluation. Downstream performance is reported at the operational quantile  $q = 0.99$ , which provides a selective but statistically stable operating point for this small-scale benchmark.

**RobustAD.** Finally, we evaluate on RobustAD [8], a recent industrial anomaly-detection benchmark designed to probe robustness under realistic distribution shifts. Here, the two typical domains used by CAP and Wasserstein are obtained by pairing nominal typical data with shifted typical data, such that our model validation and selection methods test consistency across realistic nuisance variation. Anomalous samples are again used only for downstream evaluation. Moreover, we only use the PCB subset of RobustAD, where 10% of the typical samples are used for validation and 10% for testing. Downstream performance is reported at the lower operational quantile  $q = 0.9$ . This milder quantile is used because the benchmark has substantially lower effective statistics than the LHC or CIFAR-10 settings, and more extreme tail estimates would be noisy.

## 5 Experiments

**Model families.** We evaluate several anomaly-detector families. For the LHC L1 AD dataset, these are autoencoders (AE), variational autoencoders (VAE) [28], Deep Sets (DS) [29] AEs, DS VAEs, Deep SVDD [4], and RealNVP [27]. For CIFAR-10 and RobustAD, we evaluate AE, VAE, Deep SVDD, and RealNVP. Together, these models cover deterministic, probabilistic, boundary-based, and density-based anomaly detectors. For the LHC setting, the DS are used to contrast fixed-order and permutation-invariant event representations. Further details are given in App. C.

**Checkpointing.** Each hyperparameter trial is trained for 50 epochs. For every trial, we retain three candidate checkpoints as in Fig. 1: the last checkpoint, the checkpoint maximizing an exponential moving average (EMA, decay 0.9) of  $Q'$ , and a checkpoint at which  $Q''$  stabilises, when evaluated at the quantile specific to each dataset; stabilisation is defined as less than 5% variation over 10 epochs. A single checkpoint is selected from these candidates using the comparison rule of Sec. 3.

**Hyperparameter search and trial selection.** For each model family and validation strategy, we run 600 Optuna trials using the multi-objective TPE sampler [37]. Each trial contributes one selected checkpoint and its corresponding pair of objective values: the strategy-specific primary validation quantity  $Q'$  and the secondary objective  $Q''$ . This defines a strategy-specific Pareto front over hyperparameter configurations. Examples of the CAP Pareto front for the AE and VAE, on the LHC dataset, are shown in Fig. 2. All Pareto-front trials are then retrained for 200 epochs. For the final comparison, we report the retrained Pareto-front trial with the best mean downstream anomaly-detection performance on the held-out anomaly datasets. This final oracle step is used only to summarize the best attainable downstream performance within each strategy-specific Pareto set.

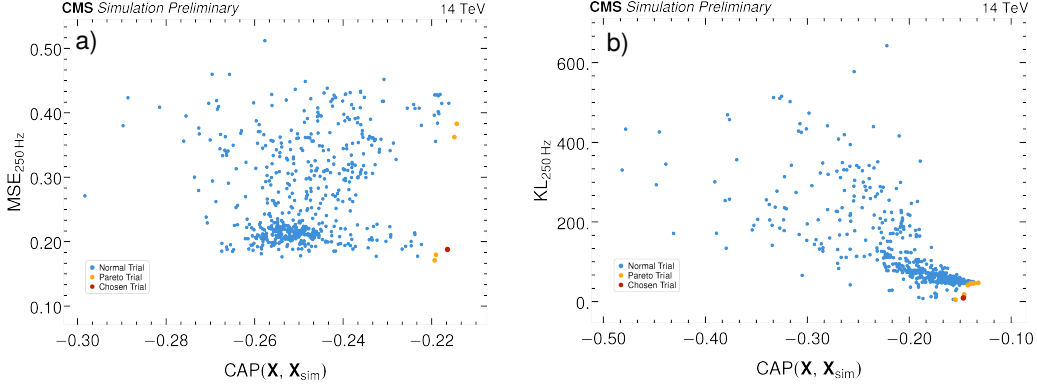


Figure 2: Pareto fronts for CAP against a) AE MSE and b) VAE KL divergence, at the relevant operating point of 250 Hz for the LHC L1 AD dataset. CAP is computed between the typical data and simulated typical data. The normal trials are coloured in blue, the Pareto front is shown in yellow, with the final selected trial being highlighted in red. This final trial is chosen as described in Sec. 5.

## 6 Discussion

**Results.** The results place CAP among the strongest signal-agnostic validation criteria considered here, with the clearest evidence on the LHC L1 AD task. Tab. 1 shows that CAP attains the highest mean downstream efficiency among the signal-agnostic methods for all six LHC model families. The most convincing gains occur for AE and DSAE, where the improvements over both stability and Wasserstein selection are large and supported by small  $p$ -values. For VAE, DSVAE, SVDD, and RealNVP, CAP remains competitive in mean performance, but the pairwise differences are smaller and often not statistically decisive. The distributional view in Fig. 3 is consistent with this pattern: AE and DSAE show a clear upward shift under CAP, while VAE, DSVAE, and RealNVP exhibit broader overlap across strategies; meanwhile, SVDD lies between these two regimes. The correlation analysis in Tab. 1 is in agreement with these conclusions. CAP is often at least as informative along training as the stability and Wasserstein-based quantities on the LHC task. The study is repeated at  $q = 0.99$  on the LHC L1 AD dataset, but the results do not change qualitatively; see App. D.

On CIFAR-10, the results are more mixed. CAP gives the best mean performance for AE and RealNVP, while Wasserstein is best for VAE and stability is best for SVDD. Among these, the AE advantage is the most clearly supported statistically; the other differences are weaker, especially for RealNVP against Wasserstein. The SVDD converges to a trivial solution on this data set for the Wasserstein-based validation paradigm, within the considered parameter space. The study is repeated using the AUPRC instead of  $q = 0.99$ : the results do not change qualitatively, see App. D.

On RobustAD, CAP again yields the best mean performance for AE and VAE and matches the top mean performance for RealNVP. At the same time, the benchmark is close to saturation for several models, which compresses the margins between strategies. Taken together, the three datasets suggest that CAP is strongest when model ranking is genuinely nontrivial and not already compressed by saturated downstream performance. The analysis is redone using the AUPRC instead of  $q = 0.9$  and the results do not change qualitatively, as shown in App. D.

**Limitations.** Several limitations remain. CAP is not uniformly strongest across all tested regimes. Additionally, it also does not exclude all high-confidence trivial solutions by itself. In the proposed pipeline, such collapse is controlled by the joint use of CAP with  $Q''$ . CAP also depends on task-specific choices: the pairing rule, the score normalization, and the energy function.

The strength of the comparison depends on the data regime. RobustAD is a very low statistics data set. In the CIFAR-10 case, the validation domains must be constructed from limited typical data, which makes the resulting quantities noisier. In contrast, for the LHC dataset, these arise naturally.

The reported model for each strategy is picked from the retrained set using downstream anomaly performance. This is an intentional choice that gives the best attainable model within each strategy, for a fair comparison, but it is distinct from the agnostic validation criterion used during search.

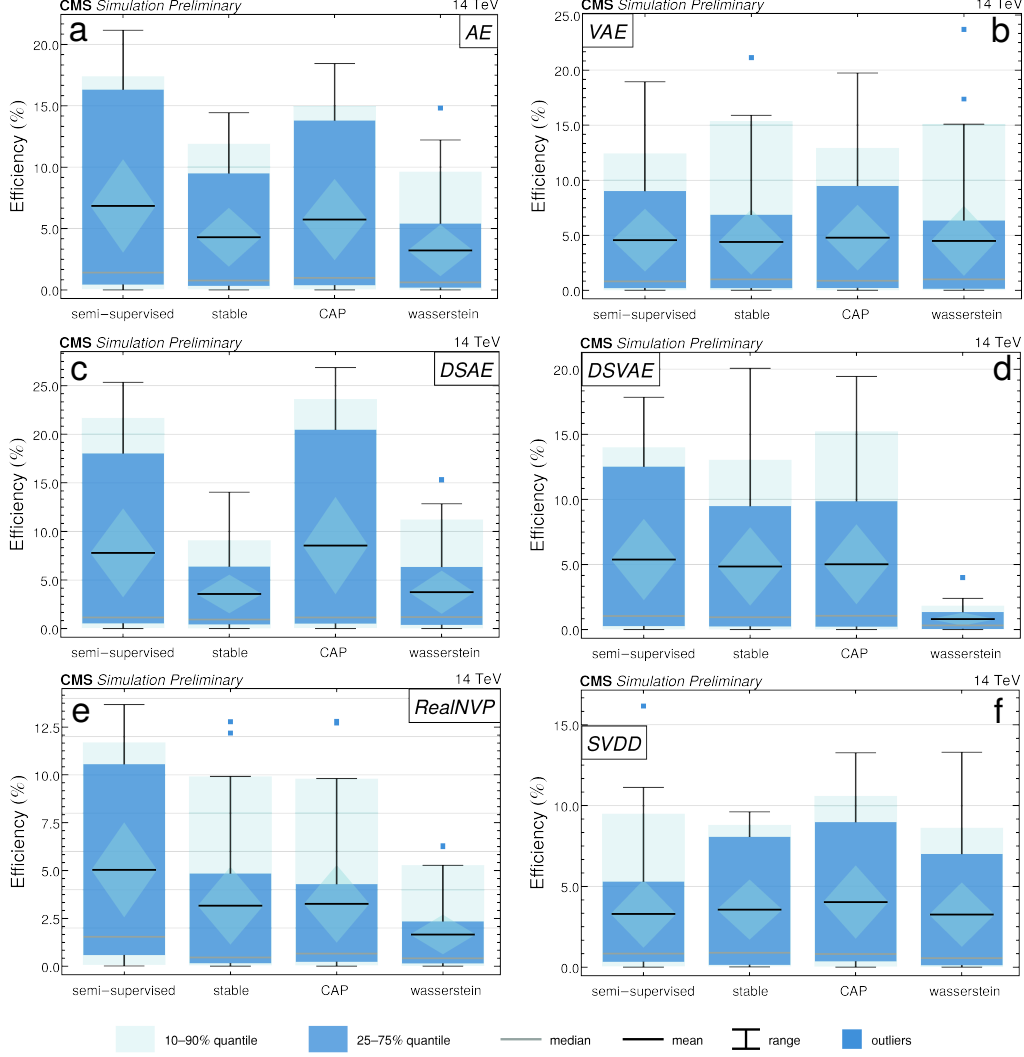


Figure 3: Distribution of downstream efficiencies, reported at the operational quantile  $q \approx 0.999991$ , over the 19 simulated physics anomaly datasets for the six models considered on the LHC L1 AD: (a) AE, (b) VAE, (c) DSAE, (d) DSVAE, (e) RealNVP, and (f) Deep SVDD. For each architecture, efficiencies are shown for all paradigms introduced in Sec. 3. The CAP model selection and validation framework is competitive with the other signal agnostic approaches, often even outperforming them. CAP dominates the other agnostic methods for reconstruction anomaly scores like AE and DSAE. In contrast, CAP does not show as clear of an improvement for the RealNVP, VAE, and DSVAE, but remains competitive for these architectures as well. The outliers depicted in the figures are determined using Tukey’s rule, i.e., they represent efficiencies that are higher than the 75th quantile +  $1.5 \times$  the 25 – 75 interquantile range.



Table 1: Mean downstream performance for the LHC L1 AD, CIFAR-10, and RobustAD studies. Semi denotes the semi-supervised reference and W1 the Wasserstein baseline. Bold marks the best signal-agnostic method for each model and dataset. The paired differences compare CAP against stability and Wasserstein; uncertainties denote bootstrap 95% confidence intervals and  $p$ -values come from paired Wilcoxon signed-rank tests. The last three columns report the Spearman correlation with median downstream efficiency, averaged over the 200 epoch training. The dashes in the table represent all the Wasserstein SVDD Pareto points giving trivial solutions on CIFAR-10.

| Model                                              | Mean (%) |               |               |               | CAP vs Stable           |       | CAP vs W1                 |         | corr w/ median |        |       |
|----------------------------------------------------|----------|---------------|---------------|---------------|-------------------------|-------|---------------------------|---------|----------------|--------|-------|
|                                                    | Semi     | CAP           | Stable        | W1            | $\Delta$                | $p$   | $\Delta$                  | $p$     | CAP            | Stable | W1    |
| <b>LHC L1 AD @ <math>q \approx 0.999991</math></b> |          |               |               |               |                         |       |                           |         |                |        |       |
| AE                                                 | 6.84     | <b>5.73</b>   | 4.29          | 3.22          | $1.44^{+0.94}_{-0.91}$  | 0.001 | $2.52^{+1.53}_{-1.38}$    | < 0.001 | 0.04           | 0.30   | 0.24  |
| VAE                                                | 4.56     | <b>4.79</b>   | 4.39          | 4.49          | $0.40^{+0.56}_{-0.55}$  | 0.263 | $0.30^{+0.75}_{-0.73}$    | 0.659   | 0.84           | -0.42  | -0.23 |
| DSAE                                               | 7.80     | <b>8.55</b>   | 3.57          | 3.75          | $4.98^{+3.54}_{-3.09}$  | 0.002 | $4.80^{+3.40}_{-2.98}$    | < 0.001 | 0.72           | 0.35   | 0.03  |
| DSVAE                                              | 5.38     | <b>5.02</b>   | 4.85          | 0.80          | $0.17^{+0.26}_{-0.20}$  | 0.172 | $4.21^{+2.47}_{-2.25}$    | < 0.001 | -0.22          | -0.31  | 0.28  |
| SVDD                                               | 3.30     | <b>4.03</b>   | 3.56          | 3.25          | $0.47^{+0.68}_{-0.65}$  | 0.121 | $0.78^{+0.46}_{-0.40}$    | < 0.001 | -0.11          | -0.51  | -0.06 |
| RealNVP                                            | 5.04     | <b>3.26</b>   | 3.17          | 1.66          | $0.10^{+0.17}_{-0.16}$  | 0.157 | $1.60^{+0.95}_{-0.83}$    | < 0.001 | 0.23           | -0.14  | -0.25 |
| <b>CIFAR-10 @ <math>q = 0.99</math></b>            |          |               |               |               |                         |       |                           |         |                |        |       |
| AE                                                 | 7.22     | <b>6.47</b>   | 5.02          | 4.72          | $1.44^{+1.13}_{-1.13}$  | 0.043 | $1.74^{+1.20}_{-1.17}$    | 0.027   | 0.73           | -0.04  | -0.15 |
| VAE                                                | 3.93     | 0.87          | 0.06          | <b>1.01</b>   | $0.81^{+0.51}_{-0.40}$  | 0.008 | $-0.14^{+0.56}_{-0.58}$   | 0.773   | 0.20           | 0.42   | 0.45  |
| SVDD                                               | 3.21     | 0.94          | <b>1.79</b>   | –             | $-0.84^{+0.34}_{-0.31}$ | 0.008 | –                         | –       | 0.14           | -0.09  | –     |
| RealNVP                                            | 9.22     | <b>5.23</b>   | 3.63          | 4.56          | $1.60^{+0.53}_{-0.57}$  | 0.008 | $0.68^{+1.64}_{-1.50}$    | 0.734   | 0.43           | -0.38  | 0.25  |
| <b>Robust AD @ <math>q = 0.9</math></b>            |          |               |               |               |                         |       |                           |         |                |        |       |
| AE                                                 | 100.00   | <b>88.65</b>  | 63.94         | 55.80         | $24.71^{+9.52}_{-9.35}$ | 0.031 | $32.85^{+11.82}_{-11.37}$ | 0.031   | -0.12          | -0.36  | -0.12 |
| VAE                                                | 92.83    | <b>40.35</b>  | 37.31         | 36.94         | $3.04^{+1.41}_{-1.48}$  | 0.063 | $3.41^{+2.04}_{-1.85}$    | 0.063   | n.c.           | -0.15  | 0.47  |
| SVDD                                               | 100.00   | 93.72         | <b>100.00</b> | <b>100.00</b> | $-6.28^{+6.28}_{-9.44}$ | 0.500 | $-6.28^{+6.28}_{-9.44}$   | 0.500   | n.c.           | n.c.   | n.c.  |
| RealNVP                                            | 100.00   | <b>100.00</b> | <b>100.00</b> | 88.57         | $0.00^{+0.00}_{-0.00}$  | 1.000 | $11.43^{+17.96}_{-10.22}$ | 0.063   | n.c.           | n.c.   | -0.14 |

n.c. denotes not computable because one of the paired series has zero variance.

## 7 Conclusion and future work

We introduced approximation capacity (CAP) by adapting the posterior agreement framework to signal-agnostic model validation for anomaly detection. Furthermore, this is achieved by specializing to anomaly detection through a binary latent hypothesis space, score-based posteriors, and a task-specific construction of the domain comparison. The resulting criterion favors models whose anomaly scores are both consistent and informative across typical data domains, going beyond stability-only model selection and validation criteria.

We also introduced a unified experimental framework in which CAP can be compared directly with semi-supervised validation, a stability criterion, and a Wasserstein distribution-similarity baseline. This puts all methods under the same checkpointing, hyperparameter-search, and retraining protocol, such that the comparison isolates the effect of the validation principle itself.

Empirically, CAP emerges as a strong signal-agnostic criterion. On the LHC L1 AD task, it gives the best mean downstream performance across all six model families and is especially effective for AE and DSAE. On CIFAR-10 and RobustAD, the picture is more mixed, but CAP remains competitive and is often strongest. Taken together, these results indicate that signal-agnostic validation can recover strong anomaly detectors without relying on labelled or synthetic anomalies, and that CAP is a useful addition to the set of such criteria.

Several future work directions remain open. The present study still relies on empirically chosen pairing, normalization, and energy functions, and the advantages of CAP are less clear in lower-statistics or near-saturated regimes. It will therefore be important to test the method on broader industrial and scientific settings. Furthermore, a natural next step is to develop more principled ways of instantiating posterior agreement criteria for anomaly detection.

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Table 2: Hyperparameter search space used in the CAP optimization study.

| Symbol                    | Meaning                                   | Values                                                                                                                                                                                                                             |
|---------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathbf{n}_{\text{enc}}$ | encoder widths<br>decoder mirrors encoder | [16, 8], [24, 8], [24, 16], [32, 8], [32, 16],<br>[32, 24], [48, 8], [48, 16], [48, 24], [24, 8, 4],<br>[24, 8, 8], [32, 16, 8], [32, 16, 16], [48, 24, 8], [48, 24, 16],<br>[48, 24, 24], [64, 32, 8], [64, 32, 16], [64, 32, 32] |
| $\sigma_{\text{in}}$      | input-noise std.                          | $\{0, 10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$                                                                                                                                                             |
| $\delta$                  | Smooth L1 shape                           | $\{0.1, 0.3, 1.0, 3.0\}$                                                                                                                                                                                                           |
| $\beta_{\text{Adam}}$     | Adam coefficients                         | $\{[0.9, 0.999], [0.9, 0.99]\}$                                                                                                                                                                                                    |
| $\eta$                    | learning rate                             | $[10^{-5}, 10^{-2}]$ (log-uniform)                                                                                                                                                                                                 |
| $\lambda_{\text{wd}}$     | weight decay                              | $\{0, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}\}$                                                                                                                                                                                        |
| $c_{\text{clip}}$         | gradient clip                             | $\{0, 0.5, 1.0, 2.0, 5.0\}$                                                                                                                                                                                                        |

## A CAP optimization

This appendix summarizes the study used to fix the CAP design choices from Sec. 3.2: the normalization  $n(x)$ , the pairing map  $P(X_1, X_2)$ , and the energy  $E(p, y)$ . The study is performed on the LHC L1 AD dataset, under the same typical-domain split and operational threshold as in the main analysis.

Three pairing rules are considered. In *empirical-CDF pairing*, scores in each domain are sorted independently and paired by rank, so that matched pairs correspond to matched score quantiles. In *absolute pairing*, samples are paired in their original order after truncation to a common length. In *random pairing*, one domain is randomly permuted before pairing. Empirical-CDF pairing is therefore the only tested rule that imposes a stable percentile-to-percentile correspondence across the two unaligned domains.

The study is carried out with an autoencoder (AE). For each tested CAP configuration, 600 hyperparameter trials are run for 50 epochs; the search space is listed in Tab. 2. The AE is trained with a Smooth L1 reconstruction loss,

$$\ell_{\delta}(r) = \begin{cases} \frac{1}{2\delta}r^2, & |r| < \delta, \\ |r| - \frac{\delta}{2}, & |r| \geq \delta, \end{cases} \quad (14)$$

where  $r$  is the reconstruction residual. The primary quantity is CAP and the secondary quantity is the mean squared error (MSE) at the operational quantile  $q \approx 0.999991$ . Within each trial, three candidate checkpoints are compared: the last checkpoint, the checkpoint maximizing the exponential moving average (EMA) of CAP, and a stable MSE checkpoint at the same quantile, defined by  $< 5\%$  variation over a 10-epoch window. These candidates are compared using the rule from Sec. 3.1. Whenever CAP is evaluated, the parameter  $\beta$  is optimized with AdamW for 20 epochs. Fig. 4 also includes one sensitivity check with initialization  $\beta_0 = 5$ ; otherwise  $\beta_0 = 1$ .

Fig. 4 shows eight representative CAP configurations. The top-left panel corresponds to the final configuration used throughout the paper: empirical-CDF pairing, sigmoid normalization, and the adaptive energy from Eq. 13. For completeness, the baseline and focal energies shown in Fig. 4 are

$$E_{\text{base}}(p, y) = y(1 - p) + (1 - y)p, \quad (15)$$

and

$$E_{\text{focal}}(p, y) = y[-\log(p + \varepsilon)(1 - p)^{\gamma}] + (1 - y)[-\log(1 - p + \varepsilon)p^{\gamma}], \quad (16)$$

where  $p \in (0, 1)$ ,  $y \in \{0, 1\}$ ,  $\gamma \equiv 2.0$ , and  $\varepsilon \equiv 10^{-8}$ , chosen for this ablation.

Among the tested variants, empirical-CDF pairing, sigmoid normalization, and adaptive energy give the most stable and informative behavior. Absolute pairing remains viable but yields a less favorable Pareto structure, while random pairing nearly collapses CAP variation. Baseline and focal energies yield weaker separation than the adaptive energy, and the shown initialization and scale variations do not improve on the selected configuration. Additional tested variants are omitted from Fig. 4 for brevity, including min-max, rank-mid, and rank normalizations; alternative pairing rules with baseline and focal energies; exponential, contrastive, and margin-based energies; further adaptive-scale variations; and an additional  $\beta$ -initialization at  $\beta_0 = 0.1$ . These variants were not retained because they did not yield comparably useful CAP behavior.

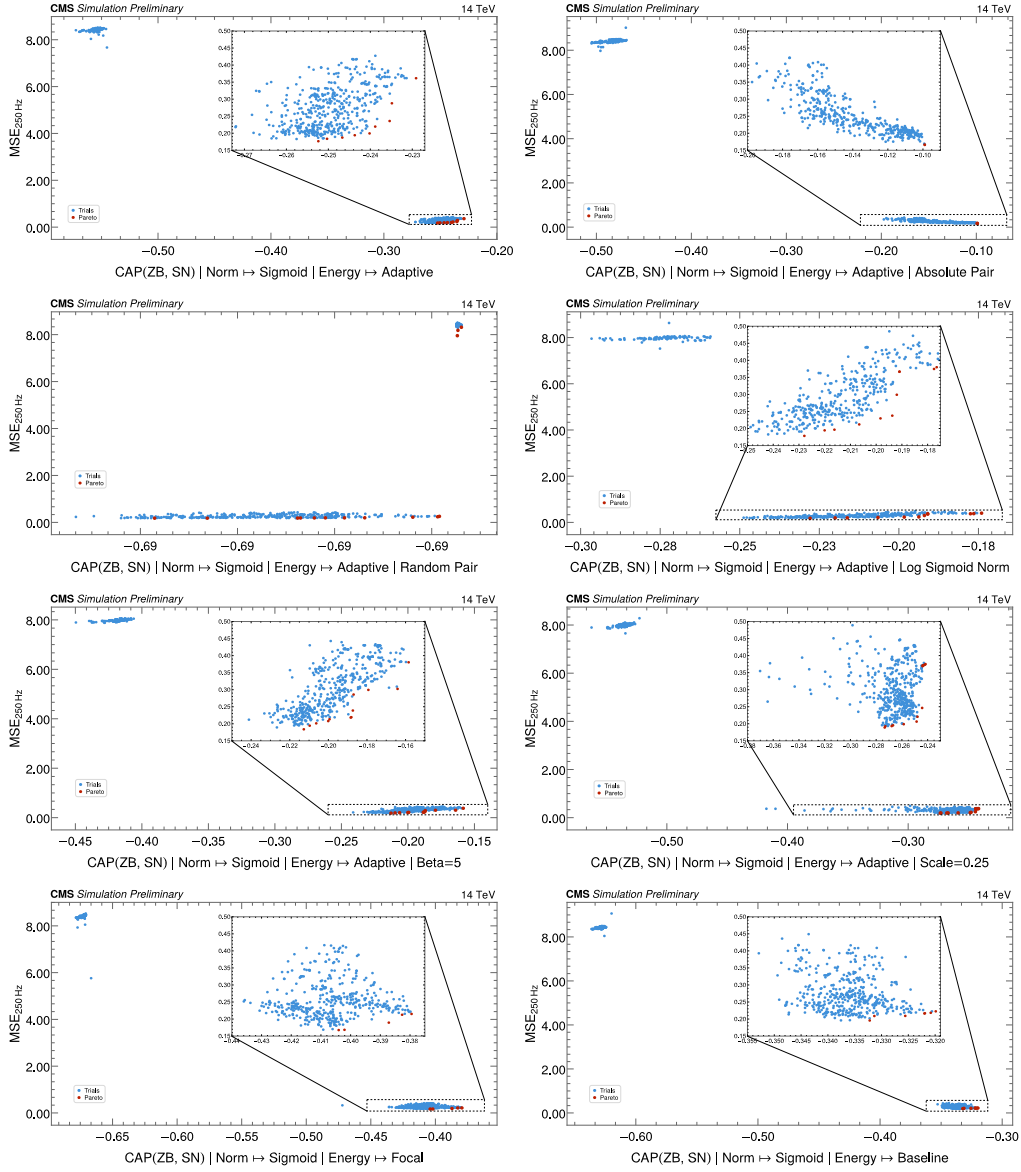


Figure 4: CAP optimization study on the LHC L1 AD dataset. Each panel shows 600 AE hyperparameter trials, with CAP as the primary quantity and MSE at the operational quantile  $q \approx 0.999991$  as the secondary quantity; red points denote Pareto-optimal trials. The displayed configurations are, from top left to bottom right: the final configuration used in the main text (empirical-CDF pairing, sigmoid normalization, adaptive energy), adaptive energy with absolute pairing, adaptive energy with random pairing, adaptive energy with log-sigmoid normalization, adaptive energy with  $\beta$ -optimization initialized at  $\beta_0 = 5$ , adaptive energy with  $\lambda = 0.25$ , focal and baseline energies. Here, ZB denotes Zero Bias data, i.e., the typical data collected in data-taking, and SN denotes Single Neutrino simulation, i.e., simulated typical data. Trials corresponding to the  $\delta < 3$  settings form the visibly poor high-MSE cluster, showing that CAP is able to capture clearly bad models.

Table 3: Composition of the LHC L1 AD dataset used in this work. ZB denotes recorded typical data and SN denotes simulated typical data.

| Category                  | Dataset                                                                       | Campaign / run           | Events     |
|---------------------------|-------------------------------------------------------------------------------|--------------------------|------------|
| Recorded typical data     | Ephemeral Zero Bias                                                           | 2025 runs 396102, 398183 | 20,087,179 |
| Simulated typical data    | Single neutrino gun ( $E = 10$ GeV)                                           | 142XWinter25             | 1,999,969  |
| Simulated anomaly samples | gluon fusion Higgs $\rightarrow b\bar{b}$ ( $m_H = 125$ )                     | 142XWinter25             | 439,006    |
|                           | gluon fusion Higgs $\rightarrow \gamma\gamma$ ( $m_H = 125$ )                 |                          | 970,566    |
|                           | gluon fusion Higgs $\rightarrow \gamma\gamma$ ( $m_H = 90$ )                  |                          | 726,841    |
|                           | gluon fusion Higgs $\rightarrow \tau\tau$ ( $m_H = 125$ )                     |                          | 59,234     |
|                           | non-standard Higgs pair production $\rightarrow 2b 2W \rightarrow \ell\nu 2q$ |                          | 291,713    |
|                           | triple-Higgs $\rightarrow 4b 2\tau$                                           |                          | 1,843,872  |
|                           | triple-Higgs $\rightarrow 6b$                                                 |                          | 588,786    |
|                           | SUSY gluon fusion $\rightarrow bbH$ , $H \rightarrow 2b$ ( $m = 1200$ )       |                          | 41,866     |
|                           | SUSY gluon fusion $\rightarrow bbH$ , $H \rightarrow bb$ ( $m = 120$ )        |                          | 499,890    |
|                           | SUSY gluon fusion $\rightarrow bbH$ , $H \rightarrow bb$ ( $m = 350$ )        |                          | 199,000    |
|                           | SUSY gluon fusion $\rightarrow bbH$ , $H \rightarrow bb$ ( $m = 600$ )        |                          | 46,346     |
|                           | top-associated Higgs $\rightarrow 2c$ ( $m_H = 125$ )                         |                          | 1,862,236  |
|                           | top-associated Higgs $\rightarrow 2b$ ( $m_H = 125$ )                         |                          | 1,868,243  |
|                           | vector-boson-fusion Higgs $\rightarrow 2c$ ( $m_H = 125$ )                    |                          | 493,394    |
|                           | vector-boson-fusion Higgs $\rightarrow 2b$ ( $m_H = 125$ )                    |                          | 494,013    |
|                           | vector-boson-fusion Higgs $\rightarrow 2\tau$ ( $m_H = 125$ )                 |                          | 1,564,196  |
|                           | $W \rightarrow \tau \rightarrow 3\mu$                                         |                          | 99,980     |
|                           | stealthy QCD-like hidden-sector decay (ggH-SUEP)                              |                          | 99,398     |
|                           | light pseudoscalar $a \rightarrow 4b$ benchmark                               |                          | 99,730     |
|                           | stealth multijet benchmark (case A)                                           |                          | 98,993     |

## 431 B LHC L1 AD dataset

432 This appendix summarizes the LHC L1 AD dataset used in the main text. The data are taken from  
433 the CMS Level-1 (L1) trigger, i.e. the hardware-based first stage of the event-selection system [38].  
434 The raw L1 ntuples are converted to parquet files using the software package in Ref. [39], which  
435 also documents the retained object and feature definitions. These parquet files are passed to the  
436 preprocessing and training pipeline implemented in Ref. [40]. The dataset consists of two typical  
437 domains together with a collection of 19 simulated anomaly datasets. The first typical domain is  
438 real recorded Zero Bias (ZB) data, i.e., collision events selected stochastically with the L1 trigger.  
439 The second is Single Neutrino (SN) simulation, which serves as a simulated proxy for the ZB data.

440 Each event is represented as a fixed collection of physics objects: muons (4), electron/photon  
441 candidates (12), jets (10), tau candidates (12), and one FET object (1). The FET object represents  
442 the missing transverse energy of an event, including CMS forward calorimeter information [38].  
443 The multiplicities correspond to the maximum number of objects of each type retained in the L1  
444 event representation; missing entries are padded with zeros. Each object is described by the features  
445 available at this stage of the detector readout, such as transverse momentum or transverse energy,  
446 angular coordinates, and quality or isolation flags. A description of the features pertaining to each  
447 object is given at Ref. [39]. In this work, only the transverse momentum, the azimuthal angle, and the  
448 pseudo-rapidity are used as features for each object.

449 Several preprocessing steps are applied before training. Events with total transverse energy above  
450 4095 GeV are removed. Individual muon, electron/photon, jet, and tau objects with transverse  
451 momentum above 511 are masked to zero, and FET objects with transverse energy above 4095 GeV  
452 are likewise masked. These values correspond to encoding limits of the hardware-level representation,  
453 so values above them are treated as overflow-like artifacts rather than physical measurements.  
454 All retained features are then robustly normalized using the 5% and 95% quantiles. This yields  
455 bounded, fixed-size event representations suitable for the model families studied in the paper.

456 The split protocol is chosen to preserve the signal-agnostic setting. Zero Bias data are divided into  
457 60% training, 20% validation, and 20% test. Single Neutrino events are divided into 60% validation  
458 and 40% test, and are not used for training. The signal simulations are divided into 60% validation  
459 and 40% test, with at most 10 batches used for signal-side validation or testing in order to control  
460 compute and class imbalance. In the main text, ZB and SN define the two typical domains used by  
461 CAP and Wasserstein validation, while the signal samples are reserved for downstream evaluation.

Table 4: Model families used in the experiments and corresponding hyperparameters.

| Model     | Meaning                                | Score type                    | Datasets | Main tuned components                                                                                                                                   |
|-----------|----------------------------------------|-------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| AE        | deterministic autoencoder              | reconstruction error          | all      | encoder widths, input noise, Smooth L1 parameter, learning rate, weight decay, gradient clipping                                                        |
| VAE       | variational autoencoder [28]           | latent KL / variational score | all      | encoder widths, latent clamp, KL scale, KL warmup, learning rate, weight decay, gradient clipping                                                       |
| DSAE      | Deep Sets autoencoder [29]             | reconstruction error          | LHC only | object-wise widths, aggregation widths, pooling, count concatenation, input noise, Smooth L1 parameter, learning rate, weight decay, gradient clipping  |
| DSVAE     | Deep Sets variational autoencoder [29] | latent KL / variational score | LHC only | object-wise widths, aggregation widths, pooling, count concatenation, latent clamp, KL scale, KL warmup, learning rate, weight decay, gradient clipping |
| Deep SVDD | one-class deep SVDD [4]                | distance to learned center    | all      | encoder widths, model weight decay, soft-boundary flag, $\nu$ , center initialization, learning rate, weight decay, gradient clipping                   |
| RealNVP   | normalizing flow [27]                  | likelihood / density score    | all      | number of coupling blocks, hidden width, hidden layers, activation, flow noise, learning rate, weight decay, gradient clipping                          |

## C Model families and Computational Resources

This appendix describes the anomaly detector models used in the experiments conducted in this work. A summary is shown in Tab. 4, together with the corresponding tuned hyperparameters.

**AE and VAE.** The autoencoder (AE) is the standard deterministic reconstruction baseline for AD. It compresses the input into a low-dimensional latent representation and is trained to reconstruct the original input, so anomalous events are expected to yield large reconstruction error. The variational autoencoder (VAE) augments this with a probabilistic latent representation and a Kullback–Leibler regularization term [28], encouraging the latent space to follow a prescribed prior. This changes both the training dynamics and the anomaly score, which can depend on latent regularity in addition to reconstruction fidelity. Together, these two families probe whether the validation conclusions depend on deterministic against probabilistic latent modeling.

**Deep Sets variants.** For the LHC L1 AD dataset, we evaluate DeepSets (DS) AEs and VAEs [29]. These models replace the standard ordered encoder with a permutation-invariant set encoder composed of object-wise feature maps followed by a symmetric aggregation operation. This is natural for the LHC event representation, where each event is more appropriately viewed as a collection of physics objects than as a fixed ordered vector. The DS variants therefore test whether the validation conclusions persist when the architecture is matched more closely to the structure of the data.

**Deep SVDD.** Deep SVDD learns a compact representation of typical data by optimizing a one-class objective around a learned center in feature space [4]. At test time, anomaly scores are given by the distance to this center, so the method seeks a representation in which typical examples occupy a compact region and anomalous examples fall outside it. This makes Deep SVDD qualitatively different from reconstruction-based models: it does not aim to reproduce the input, but instead to define a tight description of typicality in representation space. It therefore broadens the evaluation beyond autoencoding paradigms.

**RealNVP.** RealNVP is a normalizing-flow density model based on an invertible transformation between the data distribution and a simple latent base distribution [27]. Because this transformation is bijective and has tractable Jacobian determinants, the model admits exact likelihood evaluation. Anomaly scores are therefore derived from density-based quantities rather than reconstruction error or distance to a one-class center.

The full hyperparameter search space for every studied model is included in the software release associated with this publication [40]. The final selected hyperparameter values are also found therein.



Table 5: Computational resources for the hyperparameter sweeps. Each entry reports the approximate average wall-clock time per trial in minutes; values in parentheses give the corresponding aggregate GPU-hours for a 600-trial sweep. The “Conc.” column gives the number of concurrent training processes per GPU. The timings in the table correspond to the hyperparameter sweeps. Final models selected from the generated Pareto fronts were retrained separately on the CERN OpenLab cluster using NVIDIA V100 GPUs.

| Model            | GPU   | Conc. | Semi       | Stable     | W1         | CAP        |
|------------------|-------|-------|------------|------------|------------|------------|
| <b>LHC L1 AD</b> |       |       |            |            |            |            |
| AE               | L40S  | 3     | 5.0 (50)   | 5.0 (50)   | 5.5 (55)   | 40.0 (400) |
| VAE              | L40S  | 3     | 6.0 (60)   | 5.8 (58)   | 6.2 (62)   | 35.0 (350) |
| DSAE             | L40S  | 3     | 21.5 (215) | 21.5 (215) | 21.5 (215) | 65.0 (650) |
| DSVAE            | H100  | 2     | 15.0 (150) | 15.0 (150) | 16.0 (160) | 45.0 (450) |
| RealNVP          | H100  | 2     | 8.0 (80)   | 8.0 (80)   | 23.0 (230) | 37.0 (370) |
| SVDD             | L40S  | 3     | 5.1 (51)   | 5.0 (50)   | 5.3 (53)   | 32.0 (320) |
| <b>CIFAR-10</b>  |       |       |            |            |            |            |
| AE               | gh200 | 1     | 2.6 (26)   | 2.6 (26)   | 2.7 (27)   | 3.0 (30)   |
| VAE              | gh200 | 1     | 3.2 (32)   | 3.2 (32)   | 3.5 (35)   | 4.0 (40)   |
| RealNVP          | gh200 | 1     | 3.3 (33)   | 3.3 (33)   | 3.4 (34)   | 3.7 (37)   |
| SVDD             | gh200 | 1     | 2.3 (23)   | 2.3 (23)   | 2.4 (24)   | 2.7 (27)   |
| <b>RobustAD</b>  |       |       |            |            |            |            |
| AE               | L40S  | 3     | 1.0 (10)   | 1.0 (10)   | 1.0 (10)   | 1.0 (10)   |
| VAE              | L40S  | 3     | 1.2 (12)   | 1.2 (12)   | 1.2 (12)   | 1.3 (13)   |
| RealNVP          | L40S  | 3     | 10.0 (100) | 10.0 (100) | 11.0 (110) | 13.0 (130) |
| SVDD             | L40S  | 3     | 2.5 (25)   | 2.5 (25)   | 2.6 (26)   | 3.0 (30)   |

## C.1 Computational Resources and Software

Hyperparameter sweeps used 600 trials per model, dataset, and validation criterion. For the LHC L1 AD and RobustAD studies, sweeps were launched on the CERN New Generation Trigger cluster. Most NGT sweeps used NVIDIA L40S GPUs; DSVAE and RealNVP on the LHC L1 AD dataset used NVIDIA H100 GPUs. For CIFAR-10, sweeps were launched on the Swiss National Computing Center cluster using SLURM jobs on NVIDIA H200 GPUs. Unless otherwise stated, each training process used 3 CPU cores. For the LHC L1 AD study, each process used approximately 20–25 GB of host memory; for CIFAR-10 and RobustAD, each process used approximately 3 GB of host memory. Individual models used less than approximately 3 GB of GPU memory per training process. The heaviest models were the ones used for the RobustAD dataset, since it has high resolution images. Queueing time and scheduler overhead are not included in the reported timings. Tab. 5 summarizes the representative wall-clock time per trial and the corresponding aggregate compute per sweep.

The individual models selected from the generated Pareto fronts were subsequently retrained on the CERN OpenLab cluster using NVIDIA V100 GPUs. For each Pareto front, on the order of 10 selected models were retrained using a fixed random seed. Retraining used the same CPU, host-memory, and GPU-memory profile as the corresponding sweep runs. For CIFAR-10, retraining took approximately 5–10 minutes per model for AE, VAE, and SVDD, and approximately 30–40 minutes per model for RealNVP, independently of the validation criterion. For RobustAD, retraining took approximately 5–10 minutes per model for AE, VAE, and SVDD, and approximately 20 minutes per model for RealNVP. For LHC L1 AD, retraining took approximately 0.5–1.2 hours per model for Semi, Stable, and W1 selections, and approximately 3–5 hours per model for CAP selections, across model architectures. Experiments used Python  $\geq 3.10$  with PyTorch 2.11.0+cu128, TorchVision 0.26.0+cu128, PyTorch Lightning 2.6.1, TorchMetrics 1.8.2, Hydra 1.3.2, Hydra Optuna Sweeper 1.2.0, Optuna 2.10.1, MLflow 3.11.1, Datasets 4.8.4, Hugging Face Hub 1.11.0, PyArrow 22.0.0, and MPLHEP 0.4.1. The exact software environment is provided in the accompanying lock file in the software release [40].

Table 6: Mean downstream performance for the LHC L1 AD, CIFAR-10 AUPRC, and RobustAD AUPRC studies. Semi denotes the semi-supervised reference and W1 the Wasserstein baseline. Bold marks the best signal-agnostic method for each model and dataset. Paired differences compare CAP against stability and Wasserstein; uncertainties denote bootstrap 95% confidence intervals and  $p$ -values come from paired Wilcoxon signed-rank tests.

| Model                                    | Semi   | CAP           | Stable        | W1            | CAP–Stable              | $p$       | CAP–W1                  | $p$       |
|------------------------------------------|--------|---------------|---------------|---------------|-------------------------|-----------|-------------------------|-----------|
| <b>LHC L1 AD @ <math>q = 0.99</math></b> |        |               |               |               |                         |           |                         |           |
| AE                                       | 66.66  | 64.25         | <b>67.43</b>  | 61.52         | $-3.18^{+1.29}_{-1.32}$ | $< 0.001$ | $2.72^{+1.10}_{-1.18}$  | $< 0.001$ |
| VAE                                      | 72.70  | <b>73.52</b>  | 64.87         | 66.07         | $8.65^{+3.21}_{-3.06}$  | $< 0.001$ | $7.46^{+3.20}_{-2.91}$  | $< 0.001$ |
| DSAE                                     | 67.25  | 59.06         | 59.53         | <b>60.64</b>  | $-0.47^{+0.87}_{-1.00}$ | 0.436     | $-1.58^{+1.05}_{-1.21}$ | 0.006     |
| DSVAE                                    | 74.47  | <b>74.32</b>  | 63.22         | 57.21         | $11.10^{+6.52}_{-5.68}$ | $< 0.001$ | $17.12^{+4.44}_{-4.31}$ | $< 0.001$ |
| SVDD                                     | –      | 61.94         | <b>66.58</b>  | –             | $-4.64^{+2.19}_{-2.26}$ | 0.003     | –                       | –         |
| RealNVP                                  | 73.95  | <b>73.85</b>  | 36.59         | 34.65         | $1.26^{+1.35}_{-1.59}$  | 0.090     | $3.21^{+1.16}_{-1.26}$  | $< 0.001$ |
| <b>CIFAR-10 AUPRC</b>                    |        |               |               |               |                         |           |                         |           |
| AE                                       | 0.83   | <b>0.82</b>   | 0.81          | 0.81          | $0.01^{+0.01}_{-0.01}$  | 0.250     | $0.01^{+0.01}_{-0.01}$  | 0.301     |
| VAE                                      | 0.67   | 0.67          | 0.60          | <b>0.71</b>   | $0.07^{+0.02}_{-0.02}$  | 0.004     | $-0.04^{+0.03}_{-0.03}$ | 0.055     |
| SVDD                                     | 0.70   | <b>0.77</b>   | 0.67          | 0.67          | $0.09^{+0.03}_{-0.04}$  | 0.008     | $0.10^{+0.04}_{-0.04}$  | 0.008     |
| RealNVP                                  | 0.76   | <b>0.73</b>   | 0.70          | 0.70          | $0.03^{+0.01}_{-0.01}$  | 0.008     | $0.03^{+0.01}_{-0.01}$  | 0.008     |
| <b>RobustAD AUPRC</b>                    |        |               |               |               |                         |           |                         |           |
| AE                                       | 100.00 | <b>97.73</b>  | 92.41         | 90.84         | $5.32^{+2.61}_{-2.60}$  | 0.031     | $6.89^{+3.15}_{-2.71}$  | 0.031     |
| VAE                                      | 99.44  | <b>85.31</b>  | 83.93         | 83.64         | $1.38^{+0.85}_{-0.64}$  | 0.031     | $1.67^{+1.07}_{-0.83}$  | 0.031     |
| SVDD                                     | 100.00 | 99.91         | <b>100.00</b> | <b>100.00</b> | $-0.09^{+0.08}_{-0.13}$ | 0.125     | $-0.09^{+0.08}_{-0.13}$ | 0.125     |
| RealNVP                                  | 78.61  | <b>100.00</b> | 98.50         | 78.61         | $1.50^{+0.14}_{-0.18}$  | 0.031     | $21.39^{+1.69}_{-2.14}$ | 0.031     |

## D Alternative Operating Point and AUPRC Studies

This appendix provides complementary evaluations to test whether the main conclusions depend on the downstream metric or on the particular operating point chosen in the main text. For the LHC L1 AD benchmark, we repeat the efficiency study at the looser operating point  $q = 0.99$ , in contrast to the extreme-tail setting used in the main analysis. For CIFAR-10 and RobustAD, we report area under the precision–recall curve (AUPRC), which evaluates ranking quality over all thresholds rather than at a single selected point.

The results are broadly consistent with the main text. CAP remains competitive with the other signal-agnostic selectors, and its clearest gains again tend to appear for reconstruction-based models. At  $q = 0.99$ , however, the LHC L1 AD boxplots in Fig. 5 are substantially less separated than at the main operating point. The looser threshold accepts a much broader fraction of anomaly samples, which raises efficiencies across methods and compresses the differences between selection criteria. As a result, the CAP advantage visible in the extreme tail is no longer sharply resolved at this operating point. This is consistent with the interpretation that CAP is most useful in the low-rate regime, where small differences in score ordering produce large changes in accepted anomaly yield.

The AUPRC results provide a complementary threshold-integrated check on CIFAR-10 and RobustAD. Unlike the fixed-quantile studies, they measure the quality of the full ranking induced by the anomaly scores. They therefore test whether the selected models preserve useful separation between nominal and anomalous samples across thresholds, rather than only at one operating point. Across these two benchmarks, CAP remains competitive with the stability and Wasserstein baselines. This indicates that its behavior is not solely an artifact of the specific thresholds used in the main experiments, but extends more broadly to ranking quality as well.

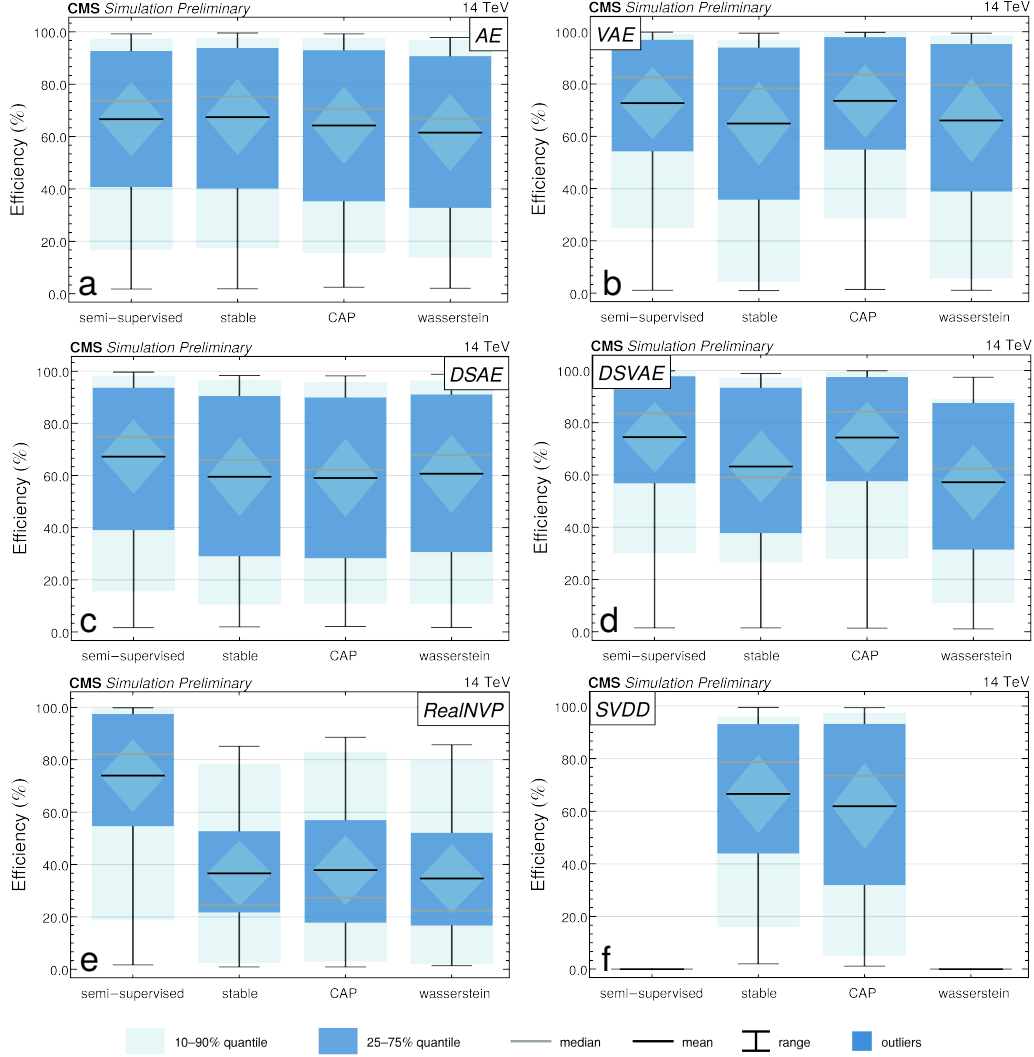


Figure 5: Distribution of downstream efficiencies, reported at the operational quantile  $q = 0.99$ , over the 19 simulated physics anomaly datasets for the six models considered on the LHC L1 AD benchmark: (a) AE, (b) VAE, (c) DSAE, (d) DSVAE, (e) RealNVP, and (f) Deep SVDD. For each architecture, efficiencies are shown for all paradigms introduced in Sec. 3. Outliers are determined using Tukey’s rule. At this looser operating point, downstream efficiencies are uniformly higher and the separation between validation criteria is more ambiguous. The Pareto SVDD models collapse for the semi-supervised and wasserstein model validation approaches.

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